Executive Summary

The `convertStPlatformToBody` module converts a star tracker attitude measurement from the sensor (case) frame into the spacecraft body frame. It takes the inertial-to-case attitude quaternion and case-frame angular velocity quaternion from the star tracker platform solution (`AscPlatformAttitudeTmMsgF32Payload`) and converts them using a configured mounting MRP.

The output is the inertial-to-body MRP $\boldsymbol{\sigma}_{B/N}$ and the body-frame angular velocity ${}^{\mathcal{B}}\boldsymbol{\omega}_{B/N}$, for downstream navigation and control modules. The sensor is rigidly mounted, so the angular velocity is only rotated into body components.

All numeric calculation is single-precision (FP32). The module has one algorithm and two interface adapters. A `SysModel` adapter connects it to the Xmera system with messages. A C shim connects it to the Adamant system through the C/Ada FFI.

Module Architecture

The module is split into a thin adapter that handles framework integration and an algorithm class that contains the pure math.

The algorithm class is `ConvertStPlatformToBodyAlgorithm`. It keeps no cycle-to-cycle measurement state. `setConfig()` computes the inverse mounting MRP (`sigma_BC`) and the case-to-body DCM (`dcm_BC`) once per configuration change.

Adapter Layer

The adapter class is `ConvertStPlatformToBody`. It inherits from `SysModel` and owns the input and output messages. Each `updateState()` copies `platformAttitude` and `platformAngularVelocity` into `q_CN` and `dq_CN` without conversion or reordering, passes them to the algorithm, and publishes `sigma_BN` and `omega_BN_B` with `solutionTimestamp` as the time tag in seconds (`double`). The output message is written every cycle. If the input message has not been written, zero quaternions and a zero time tag are used, so the output is `sigma_BC` with zero angular velocity and a zero time tag.

Table 1: Module I/O Messages

Msg Variable Name	Msg Type	Description
`stSensorInMsg`	`AscPlatformAttitude TmMsgF32Payload`	Star tracker platform attitude solution: attitude quaternion, rate quaternion and time tag.

Msg Variable Name	Msg Type	Description
`stAttOutMsg`	`STAttMsgF32Payload`	Body-frame attitude output: inertial-to-body MRP, body-frame angular velocity and the time tag passed through.

Configuration

`sigma_CB` is validated in `ConvertStPlatformToBodyConfig::create()` when the adapter's `reset()` or `reconfigure()` runs; the algorithm never sees an invalid configuration.

Table 2: Configuration parameters

Parameter Name	Type	Units	Default	Description
`sigma_CB`	`Eigen::Vector3f`	[-]	Zero	MRP of the case frame relative to the body frame. Must be finite. A shadow set is stored as its principal set. The default (zero) assumes the star tracker case frame is aligned with the spacecraft body frame; set the actual mounting MRP otherwise.

Two-Phase Initialization

The adapter is set up in two phases. First, create the module, set `sigma_CB`, connect `stSensorInMsg`, and add the module to the simulation task; nothing is checked in this phase and the algorithm is not yet built. Second, `reset()`, called by `InitializeSimulation()`, checks that `stSensorInMsg` is connected, validates `sigma_CB`, and builds the algorithm. It throws if either check fails:

```
module = convertStPlatformToBodyF32.ConvertStPlatformToBody()
module.modelTag = "stPlatformToBody"
module.sigma_CB = sigma_CB
module.stSensorInMsg.subscribeTo(stSensorMsg)

scSim.AddModelToTask(simTaskName, module)
scSim.InitializeSimulation()  # runs reset()
```

After `reset()`, `updateState()` runs each cycle. To change the mounting, set `sigma_CB` and then call `reconfigure()`; the new value is used from the next `updateState()`. If `updateState()` or `reconfigure()` is called before `reset()`, an `XmeraLifecycleException` is raised.

Mathematical Formulation

The module works with three reference frames: the inertial frame $\mathcal{N}$, the spacecraft body frame $\mathcal{B}$, and the star tracker case (platform) frame $\mathcal{C}$.

Inertial-to-Case Attitude

The star tracker gives the inertial-to-case attitude as a scalar-last quaternion $\mathbf{q}_{C/N} = [q_0, q_1, q_2, q_3]^T$, with q_3 the scalar part. It is reordered to the scalar-first Euler parameter set

$$\boldsymbol{\beta}_{C/N} = [q_3, q_0, q_1, q_2]^T$$

and converted to the equivalent MRP set $\boldsymbol{\sigma}_{C/N}$ (`epToMrp`).

Inertial-to-Body Attitude

The case-to-body MRP is the inverse of the configured mounting MRP,

$$\boldsymbol{\sigma}_{B/C} = -\boldsymbol{\sigma}_{C/B}.$$

The inertial-to-body MRP is the composition of the inertial-to-case rotation followed by the case-to-body rotation, $[BN] = [BC][CN]$, computed by MRP addition (`addMrp`):

$$\boldsymbol{\sigma}_{B/N} = \boldsymbol{\sigma}_{B/C} \oplus \boldsymbol{\sigma}_{C/N}.$$

Angular Velocity from the Rate Quaternion

The star tracker reports the case-frame angular velocity packed in quaternion form (micro-ASC ICD, ASC-DTU-ICD-3004B, Angular Velocity Quaternion Struct),

$$\delta\mathbf{q}_{C/N} = \left[\hat{\mathbf{e}}\sin\!\left(\tfrac{\|\boldsymbol{\omega}\|}{2}\right),\ \cos\!\left(\tfrac{\|\boldsymbol{\omega}\|}{2}\right)\right],$$

with $\|\boldsymbol{\omega}\|$ in rad/s. It is a rate, not a rotation over a sample interval, so no time step is needed: inverting the packing gives ${}^{\mathcal{C}}\boldsymbol{\omega}_{C/N}$ in rad/s directly. With $\delta\mathbf{q}_v = [\delta q_0, \delta q_1, \delta q_2]^T$,

$$^{\mathcal{C}}\boldsymbol{\omega}_{C/N} = \frac{2\,\mathrm{atan2}(\|\delta\mathbf{q}_v\|,\ \delta q_3)}{\|\delta\mathbf{q}_v\|}\,\delta\mathbf{q}_v.$$

The four-quadrant arctangent is used in place of arccos so that small rates (where $\delta q_3 \approx 1$) retain float32 precision. When $\|\delta\mathbf{q}_v\| = 0$ the angular velocity is set to zero, avoiding a divide-by-zero.

Body-Frame Angular Velocity

Because the sensor is rigidly mounted, $\boldsymbol{\omega}_{B/N} = \boldsymbol{\omega}_{C/N}$, so only the components change. The case-frame components are transformed into body-frame components with the transpose of the mounting DCM, $[BC] = [CB]^T$, where $[CB]$ is the DCM of the configured mounting MRP $\boldsymbol{\sigma}_{C/B}$:

$$^{\mathcal{B}}\boldsymbol{\omega}_{B/N} = [BC]\,^{\mathcal{C}}\boldsymbol{\omega}_{C/N}.$$

Assumptions and Limitations

- The star tracker case is rigidly mounted to the hub, so $\boldsymbol{\omega}_{B/N} = \boldsymbol{\omega}_{C/N}$ and `sigma_CB` is constant between reconfigurations.
- `q_CN` must have length 1, as every attitude quaternion does. The micro-ASC ICD defines it as unit length, so the module does not check or renormalize it.
- If the input message has not been written, the output is not a valid measurement and carries a zero time tag (see Adapter Layer).